Rate-Distortion Loss Attribution for AV1 Intra Partition Decisions

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***Abstract*—The AOMedia Video 1 (AV1) codec decides how to split each intra-frame super block through a recursive search that evaluates ten partition shapes per node, and learned pruners are the dominant strategy to shorten that search. Every such pruner couples the quality of the source that scores a node with the aggressiveness of the policy acting on that score, so the reported time savings do not identify which information carries the decision. This paper presents an offline attribution protocol for AV1 intra partitioning that reads the rate-distortion loss of competing representations at matched search-cost reduction. The protocol fixes one pruning policy, sweeps its confidence threshold, and charges each forced decision the overhead recorded by an exhaustive reference search, so that representations are compared at the same amount of work skipped. Experimental results over 3.81 million held-out decision nodes show that twenty-four compact descriptors reduce the loss of isolated block variance by 7.0 times, that eight causal partitioning columns reduce it by a further 1.60 times, and that four quantization and position columns add nothing. A ConvNeXt with 28.1 million parameters reading raw pixels stays 4.1 times behind the compact descriptors, which hold 2481 times fewer parameters, and doubling its fusion width makes it worse. The protocol is ordinal and does not convert into Bjontegaard delta bitrate. To the best of the authors' knowledge, this is the first work to attribute rate-distortion loss to information access in AV1 partitioning under matched cost.**

***Index Terms*—Video coding, AV1, intra-frame partitioning, feature representation, model capacity, machine learning.**

I. Introduction

Video coding removes the spatial and statistical redundancy of a raw sequence so that it can be stored and transmitted at a small fraction of its original rate. Every coding choice trades bitrate against reconstruction fidelity, and encoders resolve that trade-off by rate-distortion optimization, selecting at each decision the candidate of lowest Lagrangian cost [8]. Coding efficiency is the fidelity obtained at a given rate, and it degrades whenever a decision is taken without evaluating the candidate that the full search would have selected. That degradation, and not encoding time, is the quantity this work measures.

Two families of codecs pursue that efficiency. The standardization path produced High Efficiency Video Coding [1] and Versatile Video Coding [2], both developed jointly by ISO/IEC MPEG and ITU-T VCEG, while the Alliance for Open Media released the AOMedia Video 1 (AV1) in 2018 as an open and royalty-free codec [3], drawing on VP9, Thor, and Daala.

AV1 follows the hybrid block-based coding flow, combining intra-frame and inter-frame prediction, transforms, quantization, in-loop filters, and entropy coding, and it introduced new tools at every one of those stages [4], [5]. Among them, the partition structure was extended to ten shapes per node, against the four of VP9. That coding efficiency is paid for in computational effort, and comparative studies report AV1 encoding times far above those of its competitors [6], [7], with the gap widening at the highest quality configuration.

Inside intra-frame coding, the recursive partition search is the single most expensive stage at that configuration. The search visits a quad-tree of square nodes, evaluates up to ten shapes at each one, and descends into every node it does not resolve, so the cost grows with the number of nodes rather than with the difficulty of the content.

Several works attack this cost with learned decisions. In AV1, tool selection guided by user priority and by coding efficiency was proposed in [9], [10], a fast intra mode decision based on the accuracy of a rate-distortion model in [11], and decision trees for inter prediction in [12]. The same problem in Versatile Video Coding was addressed with light gradient boosting [13], random forests [14], and variance and gradient heuristics [15], while dedicated hardware architectures targeted the AV1 intra prediction modes [16], [17]. These works can be classified as coding-efficiency oriented [10], [11], [13]–[15], hardware oriented [16], [17], or aimed at a different coding stage [9], [12]. All of them report time savings under a policy of their own choosing, and none of these works separates the quality of the scoring source from the aggressiveness of the policy, nor measures which information is decisive for the partition choice.

This paper presents an offline attribution protocol for AV1 intra partitioning. The main strategy of this work is to hold the pruning policy and the amount of skipped search work fixed, and to vary only what a representation is allowed to see, so that the resulting difference in rate-distortion loss is attributable to information access alone. Under this protocol, compact descriptors outperform a high-capacity convolutional model reading raw pixels, and the causal partitioning neighborhood carries decision information that luminance does not provide.

II. Intra Partition Search in AV1

AV1 partitions each frame into super blocks of 128×128 samples, which the encoder may alternatively set to 64×64 [4]. Each super block is recursively divided by a search over a quad-tree of square nodes that may descend to 4×4 samples. At every node the encoder may evaluate ten partition shapes, organized in four groups: the undivided block, the quaternary split, the two rectangular shapes, and the six extended shapes.

The evaluation order inside a node is fixed. The undivided candidate is evaluated first and produces a complete rate-distortion cost; the quaternary split follows, and opening it triggers the recursion into the four children; the rectangular shapes come next, and the extended shapes last.

This order defines when each kind of information becomes available, the property this work exploits. Before the search of a node begins, the encoder already holds the source luminance of the block, the frame quantization index, the position of the node, and the partition decisions of the causal neighbors above and to the left, but it does not yet hold any rate-distortion cost for the current node. Fig. 1 summarizes this timeline.

The action studied in this paper is the earliest one available: committing the node to the undivided shape before any rate-distortion evaluation is paid, and therefore skipping every remaining candidate of the node and the whole subtree below it. This single action has two measurable consequences, the search work that is no longer performed and the rate-distortion cost that is lost whenever the discarded subtree was the optimal choice.

This work is focused on that commitment at the pre-search insertion point of the intra-frame partition tree. Inter-frame prediction, transform type selection, quantization, in-loop filters, and entropy coding are not included in the loop attacked here, and therefore they are not focused on this work. The decision at the 128×128 root is also outside this scope: the protocol attaches at the four 64×64 quadrants of each super block and models the 64, 32, and 16 sample levels. Blocks of 8×8 samples were never split in this corpus, no divided label appearing among 16.91 million of them, so 4×4 nodes do not occur and 8×8 is the effective leaf; they are excluded from modeling while kept in the cost accounting.

III. Proposed Attribution Protocol

The proposed protocol compares representations by what they are allowed to see, under one fixed pruning policy and at a matched amount of skipped search work.

PLACEHOLDER — information-availability timeline inside one node: causal context and source luminance available before the search; undivided cost available after the first candidate; subtree costs available only after the recursion. The pre-search instant is marked.

Fig. 1. Information available to a pruner at each instant of the search of a partition node, and the pre-search instant addressed in this work.

A. Representations Declared by Information Access

The representations are declared by information access rather than by architecture. The first is the isolated variance of the block, the classical descriptor of partition heuristics. The second, denoted A, is a twenty-four column vector of compact luminance descriptors: second-order and gradient statistics of the block and of its quadrants, edge measures, and the contrast against its parent and sibling blocks. Three of those columns do not derive from luminance, namely the normalized quantization index and the two coordinates of the node inside the 64-sample unit.

Block B adds eight columns of causal partitioning neighborhood: availability of the above and left neighbors, their already decided widths and heights in logarithmic scale, the relative granularity of the neighborhood, and its anisotropy. Block C adds four columns of effective quantization and position, namely the dequantization step of the direct-current coefficient, the normalized position within the frame, and the depth of the node. Four rungs are built from these blocks, namely A, A+B, A+C, and A+B+C, the last one holding thirty-six columns. All of these columns are natively available during encoding at the pre-search instant, ensuring zero computational overhead for feature extraction.

The remaining representation is the deep pixel arm, a multi-level convolutional model of the ConvNeXt family [18] that mirrors the partition hierarchy. Its input is a two-channel tensor formed by the luminance of the 64×64 unit and a constant quantization plane. Three stages of its trunk are merged top-down, so that each cell of the resulting map holds both local detail and the context of the whole unit, and per-level heads read average poolings of it. This arm is present as a test of the hypothesis that raw pixels plus representation capacity suffice, and not as an architectural baseline. A random scorer is included as a control.

The four rungs are multilayer perceptrons with two hidden layers of 64 and 32 rectified units, one network per block size, trained with hard-label cross entropy and AdamW [20], learning rate 1×10^{-3}, batches of 4096, and a fixed budget of 30 epochs. Every rung is trained with this one recipe and differs only in which columns it reads, which is what makes the verdict fall on information rather than on capacity or on training procedure. Each rung is trained with three seeds and the reported value is their mean.

B. Policy, Cost Model, and Loss

The policy is fixed for every representation. A node n is committed to the undivided shape whenever the score assigned to that class exceeds a confidence threshold τ, in which case the node contributes only the cost of its undivided candidate and the recursion stops; otherwise the node is evaluated in full and the search descends. The rate-distortion loss charged to a committed node is the overhead of that commitment against the optimal subtree, as recorded by the exhaustive reference search,

$$r(n) = J_{\mathrm{none}}(n) - J^{*}(n), \tag{1}$$

where $J_{\mathrm{none}}(n)$ is the rate-distortion cost of the undivided candidate and $J^{*}(n)$ is the cost of the optimal subtree rooted at n. It is non-negative by construction, and zero whenever the undivided shape already was the reference decision. No encoding is repeated, since the whole protocol is a counterfactual replay over the log of one exhaustive search.

The search work of a node of dimension n is counted analytically as

$$c(n) = k_n\, n^2, \quad k_n = 9 \text{ for } n \in \{64, 32, 16\},\ k_8 = 4, \tag{2}$$

that is, the number of shape candidates a full search would evaluate, weighted by the area of the block. The cost reduction $\Delta(\tau)$ is the percentage difference between the work performed by the policy, accumulated with (2), and the work of the full search. It is a counted quantity and not wall-clock time, nor an estimate of it.

Summing (1) over the committed nodes gives the loss the pruning incurred,

$$L(\tau) = 100\, \frac{\sum_{n \in P(\tau)} r(n)}{\sum_{n \in N} J_{\mathrm{none}}(n)}, \tag{3}$$

where $P(\tau)$ is the set of nodes committed at threshold τ and N is the set of all decision nodes. The denominator is accumulated before any threshold is fixed and stays constant along the sweep, so L tends to zero as pruning tends to zero.

Lowering τ prunes more, which raises Δ and L together. Each representation is therefore a curve rather than a point, and representations are compared by reading their curves at the same cost reduction. The reference reading in this paper is $\Delta = 25\%$, one quarter of the shape evaluations skipped.

Two properties of L restrict its reading, and they are stated before any number is reported. First, the denominator aggregates the undivided costs of the three hierarchy levels, which cover the same image area, so it is systematically larger than the cost of coding the frame and L is systematically smaller than any efficiency loss observable in the encoder. Second, the replay discounts recorded values, whereas pruning inside the encoder also alters the reconstruction that serves as reference to later blocks, a propagation absent here. Consequently, no conversion factor between L and the Bjøntegaard delta bitrate [25] is postulated, and L is used strictly in an ordinal regime, in which every ratio is taken between representations read at the same operating point.

C. Dataset and Experimental Setup

The reference labels were extracted from libaom v3.10.0 [22], instrumented under a compile-time guard so that the production binary remains byte-identical to the original. The encoder was configured for All-Intra following the AOM Common Test Conditions [23] at `cpu-used=0`, which enables the full rate-distortion search and therefore makes the recorded decisions optimal by construction.

The corpus comprises sixteen sequences of the UVG dataset [24] at 4K resolution (3840×2160), 8 bits and 4:2:0 chroma subsampling. The restriction to 4K is deliberate. The search cost grows with the number of nodes, so this is the resolution at which the exhaustive partition search is most prohibitive; and resolution also governs the super block size the encoder selects, so averaging over resolutions would average over different tree depths. Each sequence was encoded at four quantization points (`cq-level` 20, 32, 43, and 55) over five frames sampled uniformly along the whole clip, since consecutive frames are nearly identical under All-Intra. The resulting dataset holds 26.98 million samples, of which 10.07 million are decision nodes. The split is by sequence and was frozen before any measurement: ten sequences for training, three for validation, three for the reserved test set. No video sequence was reused across the training, testing, or evaluation datasets, thus avoiding data leakage and overfitting. All results reported here are measured on the six held-out sequences, which contribute 226 447 units of 64 samples and 3 808 703 decision nodes. The models were implemented in PyTorch [21].

D. Fair Treatment of the Deep Arm

The deep arm carries the hypothesis that raw pixels plus capacity suffice, so a negative result on it is only meaningful if the arm was given every advantage. Three measures were taken.

The first concerns the corpus. An audit of an earlier checkpoint revealed that this arm had been trained on a much smaller set, at a single quantization point and with two of the six held-out sequences present in training; that comparison was discarded, and the arm was retrained on the same corpus and the same split as the rungs. The second concerns capacity: the fusion width was doubled from 128 to 256 channels, and the wider model turned out 1.0% worse in validation loss, so capacity is not the binding constraint.

The third concerns the training objective. A plain cross entropy and a cost-sensitive variant [19], which weights each node by the rate-distortion loss its misclassification would incur, were trained under identical corpus, schedule, and selection criterion. The cost-sensitive objective proved better at every operating point from 10% of cost reduction onward, and it is the deep result discussed in Section IV. This inverts a conclusion previously drawn by the authors from an uncontrolled comparison, so the value reported for the deep arm is the most favorable one measured.

TABLE I
RATE-DISTORTION LOSS AT MATCHED COST REDUCTION, IN UNITS OF 10^{-3} PERCENT OF THE AGGREGATED UNDIVIDED COST

Representation	**10%**	**15%**	**20%**	**25%**	**30%**
Random control	196.3	298.9	406.6	516.6	634.2
Variance	4.3	12.7	25.0	37.4	55.5
ConvNeXt, plain CE	6.5	10.5	16.4	27.2	44.2
ConvNeXt, width 256	7.5	12.2	19.4	29.4	44.4
ConvNeXt, cost-sensitive	5.4	8.0	14.0	21.9	37.9
A (24 columns)	0.5	1.3	2.8	5.3	9.1
A+C (28 columns)	0.5	1.2	2.7	6.4	12.2
A+B+C (36 columns)	0.2	0.6	1.8	4.0	7.8
A+B (32 columns)	**0.2**	**0.5**	**1.5**	**3.3**	**6.9**

PLACEHOLDER — rate-distortion loss versus matched cost reduction, logarithmic vertical axis, one curve per representation, with seed-amplitude bars on the four rungs.

Fig. 2. Rate-distortion loss as a function of matched search-cost reduction. Bars on the rungs show the amplitude over three training seeds.

IV. RESULTS AND DISCUSSION

The representations were evaluated over the six held-out sequences and 3 808 703 decision nodes, under the All-Intra configuration of libaom v3.10.0 at `cpu-used=0`. Table I reports the loss L of (3) at five matched cost reductions, in units of 10^{-3} percent, where lower is better, and Fig. 2 shows the corresponding frontiers.

The main conclusion when observing Table I is that the ordering of the representations is preserved across the whole range, so it is not an artifact of one operating point. At 25%, isolated variance loses 37.4 units while the twenty-four compact columns of rung A lose 5.3, a reduction of 7.0 times, against the 516.6 units of the random control. This is the largest single gain in the table, and it contradicts any reading in which the information available in the pixel domain would be exhausted by variance.

Adding the eight causal partitioning columns to rung A lowers the loss from 5.3 to 3.3 units, a further reduction of 1.60 times. The separation is robust to initialization, since the worst of the three seeds of A+B, at 3.6, is still better than the best of the three seeds of A, at 4.9. This is the central result: those eight columns describe the partition choices of the already coded neighbors, are not computable from the luminance of the block under any transformation, and therefore carry information produced by the encoding process itself.

Conversely, the four columns of effective quantization, frame position, and node depth add nothing. Rung A+C reaches 6.4 units against 5.3 for rung A at 25%, and its seed amplitude is 4.5 units against 0.8 for rung A, so the two distributions overlap and block C does not add information while destabilizing the fit. Once combined with the causal columns the effect is no longer ambiguous, as A+B reaches 3.3 units against 4.0 for A+B+C, all nine pairwise seed comparisons agree, and Fig. 2 shows A+B dominating A+B+C across the entire frontier. The thirty-six column vector is therefore not the offline optimum, although this superiority of A+B has not been verified inside the encoder.

The deep arm does not support the hypothesis that raw pixels plus representation capacity suffice. In its most favorable configuration it reaches 21.9 units at 25%, which is 4.1 times worse than rung A, although it holds 28 128 638 parameters against the 11 337 of the three networks of rung A, a factor of 2481. Doubling the fusion width degrades the result rather than improving it, at 29.4 against 27.2 units under the same objective. This may be attributed to missing information rather than to available capacity: the deep arm reads the same pixels as rung A, and eight columns of encoder state close a further factor of 1.60 with 4291 parameters per level. The deep arm is therefore not an upper bound of the pixel domain, since a model outperformed by another with strictly smaller information access establishes no ceiling.

Four limitations qualify these results and are reported as part of them. The protocol is ordinal, as established in Section III-B, so none of these values converts into a Bjøntegaard delta bitrate or predicts what an encoder would deliver. The three seeds provide an amplitude and not a confidence interval. The objective ablation of the deep arm covers two objectives, and the strength of the cost weighting was not swept. Finally, the corpus is 4K only, so whether the causal neighborhood carries the same weight where the super block is 64×64, and the tree is one level shallower, is not established here.

V. CONCLUSIONS

This paper presented an offline attribution protocol that measures the rate-distortion loss of competing representations of the AV1 intra partition decision at matched search-cost reduction, reaching the conclusion that the causal state of the encoding process carries decision information that luminance alone does not provide. The protocol holds the pruning policy and the training recipe fixed and varies only the information a representation may read. It was evaluated over 3.81 million held-out decision nodes of sixteen UVG 4K sequences encoded with libaom v3.10.0. Twenty-four compact descriptors improve on isolated variance by 7.0 times, eight causal partitioning columns improve on those by a further 1.60 times, and four quantization and position columns add nothing. A ConvNeXt with 28.1 million parameters remains 4.1 times behind the compact descriptors, and doubling its width makes it worse, so capacity is not the binding constraint. Besides, to the best of the authors' knowledge, this is the first attribution of rate-distortion loss to information access in AV1 partitioning.

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